Intellectual Developmental Disorder With Macrocephaly, Seizures and Speech Delay (PAK1-related disorder)

MONDO:0032568 · OMIM:618158 · Causal gene: *PAK1* (p21-activated kinase 1, HGNC:8590)

Summary

Intellectual Developmental Disorder with Macrocephaly, Seizures and Speech Delay (IDDMSSD; MONDO:0032568, OMIM:618158) is an **ultra-rare, autosomal-dominant neurodevelopmental disorder caused by de novo, gain-of-function missense variants in *PAK1***, the gene encoding p21-activated kinase 1 (HGNC:8590; UniProt Q13153). Despite the serial-number style of the disease name, the causal gene is unambiguously **PAK1** — a serine/threonine kinase that acts as a direct downstream effector of the Rho-family small GTPases Cdc42 and Rac1 and regulates actin cytoskeletal dynamics in developing neurons. To date only ~16–20 patients have been reported worldwide, and every single disease-associated variant reported has been **missense** — there are no truncating/loss-of-function patients — which, together with population constraint data, is the central genetic signature of a gain-of-function mechanism rather than haploinsufficiency.

Mechanistically, resting PAK1 exists as a **trans-autoinhibited homodimer** in which the N-terminal autoregulatory region of one protomer binds and silences the kinase domain of the other. Binding of GTP-loaded Cdc42/Rac1 dissociates the dimer and activates the kinase. Disease variants cluster in two hotspots — the **N-terminal autoregulatory/CRIB region (~aa 70–140)** and the **C-terminal protein kinase domain (aa 270–521)** — and both classes relieve autoinhibition, promoting dimer dissociation, increased autophosphorylation, and constitutive kinase activity. The hyperactive kinase dysregulates the **Rac1/Cdc42 → PAK1 → LIMK1 → cofilin** actin-remodeling axis that controls neurite outgrowth, dendritic spine morphology, synaptic plasticity, and ultimately postnatal brain size.

Clinically the disorder produces moderate-to-severe intellectual disability, speech delay, (often postnatal) macrocephaly/megalencephaly with ventriculomegaly, seizures (ranging from febrile and focal seizures to infantile epileptic spasms syndrome), hypotonia, autistic features, structural brain anomalies, and dysmorphism. A domain-dependent genotype–phenotype correlation has been described: autoregulatory-domain variants associate more with neuroanatomical alterations, whereas kinase-domain variants associate more with non-neurological comorbidities. The disease course is a **static (non-progressive) encephalopathy**; there is **no approved disease-modifying therapy**, and management is entirely symptomatic (anti-seizure medication, hydrocephalus shunting, and developmental/speech/physical/occupational therapies). PAK1 small-molecule inhibitors are a mechanistically rational but still preclinical precision-treatment prospect.

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1. Disease Information

Overview. IDDMSSD is a Mendelian, monogenic neurodevelopmental disorder defined by the triad embedded in its name — intellectual disability, macrocephaly, and speech delay — plus seizures, caused by heterozygous activating missense variants in *PAK1*.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0032568
OMIM	618158
Gene (HGNC)	PAK1, HGNC:8590
Gene (NCBI)	5058
Ensembl	ENSG00000149269
UniProt	Q13153 (PAK1_HUMAN)
Inheritance	Autosomal dominant, de novo

Synonyms / alternative names: "PAK1-related neurodevelopmental disorder"; "neurodevelopmental disorder with macrocephaly, seizures, and speech delay"; "intellectual disability with macrocephaly and seizures." Orphanet, ICD-10/ICD-11, and MeSH do not currently carry a dedicated, specific term for this ultra-rare entity; coding is generally via generic categories for intellectual disability/developmental disorder.

Source of information. The knowledge base is derived from **aggregated disease-level resources and individual published case reports / small case series** (trio exome/genome sequencing studies), not from large EHR cohorts — reflecting the ultra-rarity of the condition.

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2. Etiology

Primary cause. The disorder is **purely genetic** — caused by heterozygous **de novo gain-of-function missense variants in *PAK1***. There is no known environmental, infectious, or multifactorial contribution.

- Horn et al. 2019 (PMID: 31504246) identified de novo heterozygous missense variants in PAK1 in four unrelated individuals: *"we identified de novo heterozygous missense variants in PAK1 in four unrelated individuals with intellectual disability, macrocephaly and seizures."*

Genetic risk factors. The causal variants are themselves the risk — they arise de novo in the affected child; unaffected parents carry no risk allele. No susceptibility loci or modifier genes have been established. One report described a patient carrying both a de novo likely-pathogenic PAK1 variant (p.T84R) **and** a de novo terminal 1q44 microdeletion, but the authors attributed the main clinical features to the PAK1 variant (PMID: 36624491).

Environmental / lifestyle risk factors. None identified or expected for a de novo monogenic disorder. Advanced parental age is a general (non-specific) correlate of de novo mutation rate but has not been specifically implicated here.

Protective factors. None known (genetic or environmental).

Gene–environment interactions. None documented. Not applicable to the current understanding of this disorder.

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3. Phenotypes

The phenotype spans a spectrum from mild developmental delay with normal head size to severe intellectual disability with refractory epilepsy and spastic quadriplegia. Core and associated features, with suggested HPO terms:

Phenotype	HPO term	Type	Frequency / notes
Intellectual disability	HP:0001249	Clinical sign	Moderate–severe; core feature
Macrocephaly / megalencephaly	HP:0000256 / HP:0001355	Physical	~4/5 cases; often **postnatal** onset
Speech delay	HP:0000750	Clinical sign	Core feature
Seizures	HP:0001250	Clinical sign	~4/5 cases; focal, GTC, epileptic spasms
Infantile epileptic spasms (IESS)	HP:0012469 / HP:0011097	Clinical sign	Severe end of spectrum
Global developmental delay	HP:0001263	Clinical sign	Common
Ventriculomegaly	HP:0002119	Imaging	Associated with macrocephaly
Hypotonia	HP:0001252	Clinical sign	Common
Autistic behavior	HP:0000729	Behavioral	Autism-like features reported
Structural brain anomalies	HP:0012443	Imaging	Common
Dysmorphic features	HP:0001999	Physical	Common
Hydrocephalus (obstructive)	HP:0000238	Imaging/ clinical	Severe cases
Spastic quadriplegia	HP:0002510	Clinical sign	Severe cases
White matter hyperintensities	HP:0002500	Imaging	Severe cases
Horseshoe kidney	HP:0000085	Physical	Reported comorbidity (kinase-domain case)

Phenotype characteristics. *Age of onset:* congenital to infancy for developmental features; macrocephaly is frequently postnatal and progressive in infancy. *Severity:* variable (mild DD → severe ID). *Progression:* the encephalopathy itself is **static/non-progressive**, though epilepsy

may be medically refractory. *Frequency:* in the combined Wang 2025 series (n=5) plus literature (n=11), **4/5 had epilepsy and 4/5 had macrocephaly**; one patient had only mild DD with normal head circumference (PMID: 41124873).

- *"Four of five patients with PAK1 variants had epilepsy, the seizure types included focal seizures, generalized tonic-clonic seizure and epileptic spasms"* (PMID: 41124873).
- Corriveau et al. 2023: *"other common characteristics include structural brain anomalies, delayed development, hypotonia, and dysmorphic features"* (PMID: 36905087).

Quality-of-life impact. Substantial and lifelong: intellectual disability and speech delay impair communication, education, and independence; refractory epilepsy and (in severe cases) spastic quadriplegia and hydrocephalus add major care burdens. No disease-specific EQ-5D/SF-36 data exist for this ultra-rare condition.

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4. Genetic / Molecular Information

Causal gene. *PAK1* (p21-activated kinase 1), HGNC:8590, NCBI Gene 5058, Ensembl ENSG00000149269, chr11:77,322,017–77,474,635 (GRCh38); 545-aa protein (UniProt Q13153).

Pathogenic variants. All reported disease variants are **heterozygous, de novo, missense**. Across Wang 2025 (n=5) and prior literature (n=11), **all 16 reported variants were missense** (PMID: 41124873: *"in our study (n = 5) and published studies (n = 11), all variants were missense variants"*).

Representative variants and their domains:

Variant (protein)	cDNA	Domain	Source
p.Thr84Arg	c.251C>G	Autoregulatory/CRIB	PMID: 36624491
(variants near autoinhibitory switch)	—	Autoregulatory	PMID: 31504246
p.Ile312Ser	—	Kinase domain	PMID: 41124873
p.Asp407Asn	—	Kinase domain (milder)	PMID: 41124873
p.Met453Thr	—	Kinase domain	PMID: 41124873
p.Leu470Pro (mosaic, VAF 13.4%)	—	Kinase domain	PMID: 41124873
p.Ile476Thr	—	Kinase domain	PMID: 41124873
p.Leu470Gln	c.1409T>A	Kinase domain (recurrent residue)	PMID: 36905087

Variant classification (ACMG/AMP): reported variants are classified pathogenic / likely pathogenic, supported by de novo occurrence, absence from population databases, clustering in functional domains, and functional gain-of-function evidence.

Allele frequency. Causal variants are **absent from gnomAD/population databases** (de novo, private).

Somatic vs germline. Variants are germline/constitutional de novo; **post-zygotic mosaicism** has been documented (p.Leu470Pro, VAF 13.4%), relevant to recurrence counseling (PMID: 41124873).

Functional consequence — gain of function. Horn 2019 showed the variants *"alleviated dissociation into monomers, autophosphorylation and activation of PAK1 influences the actin dynamics of neurite outgrowth"* (PMID: 31504246) — i.e., constitutive kinase activation.

Population constraint (gnomAD v2). PAK1 is **strongly missense-constrained but LoF-tolerant**, exactly the signature of a missense gain-of-function gene:

Metric	Value	Interpretation
Missense Z	**5.50**	Extreme missense intolerance
Observed/expected missense	0.50	Half the expected missense seen
pLI	0.069	LoF-tolerant
LOEUF (oe_lof upper)	0.587	Relatively LoF-tolerant
lof_z	3.97	—

This pattern (high mis_Z, low pLI) argues against haploinsufficiency and for a pathogenic mechanism acting through altered (hyperactive) protein — consistent with the all-missense variant spectrum.

Modifier genes / epigenetics / chromosomal abnormalities. No established modifier genes. No disease-specific epigenetic signature reported. One patient had a concurrent de novo 1q44 microdeletion, but phenotype was attributed to PAK1 (PMID: 36624491).

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5. Environmental Information

Not applicable. This is a de novo monogenic disorder with **no known environmental, lifestyle, or infectious contributing factors**. No toxins, radiation, occupational exposures, dietary factors, or pathogens have been implicated.

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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **de novo missense variant** arises in *PAK1*, in either the N-terminal autoregulatory/CRIB region (~aa 70–140) or the kinase domain (aa 270–521). → *leads to*
2. **Disruption of the trans-autoinhibited PAK1 homodimer.** Normally the N-terminal inhibitory segment of one protomer binds and silences the catalytic domain of the other; the variant weakens this interface. → *results in*

3. **Alleviated dimer dissociation into monomers and increased autophosphorylation** → *leads to*
4. **Constitutive, GTPase-independent kinase hyperactivation** (gain of function) → *leads to*
5. **Dysregulated downstream actin signaling** via the LIMK1 → cofilin axis (and other PAK1 substrates), altering actin filament dynamics → *results in*
6. **Abnormal neurite outgrowth, dendritic spine morphology, and synaptic properties** in developing neurons → *leads to* (branch)
7. 6a. **Aberrant postnatal brain growth / megalencephaly** (brain size is set by PAK1-dependent neuronal complexity) → macrocephaly/ventriculomegaly
8. 6b. **Disrupted synaptic plasticity and cortical circuit formation** → intellectual disability, speech delay, autism features
9. 6c. **Cortical hyperexcitability / network dysregulation** (inferred) → seizures, up to infantile epileptic spasms
10. The combined circuit-level defects manifest as the **clinical phenotype**: ID, macrocephaly, speech delay, seizures, hypotonia, structural brain anomalies.

> Steps 2–5 are directly demonstrated in vitro (Horn 2019; Parrini 2002). Step 6/brain-size link is supported by mouse loss-of-function genetics (Huang 2011). The specific epileptogenic mechanism (6c) is **inferred**, not yet directly demonstrated for patient variants.

Detail

Molecular pathways. PAK1 is a direct effector of the **Rho-family GTPases Cdc42 and Rac1**. The disease axis is **Rac1/Cdc42 → PAK1 → LIMK1 → cofilin** , governing actin cytoskeletal remodeling. KEGG/Reactome pathways: Rho GTPase signaling, axon guidance, regulation of actin cytoskeleton. Parrini et al. 2002 established the structural basis: *"The dimerized Pak1 adopts a trans-inhibited conformation: the N-terminal inhibitory portion of one Pak1 molecule in the dimer binds and inhibits the catalytic domain of the other"* and the dimer dissociates upon GTP-Cdc42/Rac1 binding (PMID: 11804587).

Cellular processes. Actin filament dynamics, neurite/axon/dendrite outgrowth, dendritic spine formation, synaptic plasticity, and control of neuronal soma size and complexity. GO terms: GO:0030036 (actin cytoskeleton organization), GO:0007409 (axonogenesis), GO:0050773 (regulation of dendrite development), GO:0048812 (neuron projection morphogenesis), GO:0050808 (synapse organization), GO:0006468 (protein phosphorylation).

Protein dysfunction. Gain of function via **loss of autoinhibition** → constitutive kinase activity (not misfolding/aggregation/LoF). Catalytic active site Asp389; CRIB/GTPase-binding motif aa 75–105; autoregulatory region aa 70–140; protein kinase domain aa 270–521.

Brain-size control — model evidence. Huang et al. 2011 showed PAK1 (with PAK3) controls postnatal brain size through neuronal complexity and cofilin-actin dynamics: *"double-knockout (DK) mice lacking both PAK1 and PAK3 were born healthy, with normal brain size and structure, but severely impaired in postnatal brain growth, resulting in a dramatic reduction in brain volume"* (PMID: 21115725). LoF reduces brain size; the patient GoF variants push in the opposite direction, consistent with **megalencephaly**.

Related-disorder corroboration. Aberrant Rac1-cofilin signaling drives dendritic-spine and synaptic defects in fragile X syndrome, and PAK inhibition rescues such defects — establishing the Rac-PAK-cofilin axis as pathogenic in neurodevelopmental disease and providing the rationale for PAK inhibition: *"Aberrant Rac1-cofilin signaling mediates defects in dendritic spines, synaptic function, and sensory perception in fragile X syndrome"* (PMID: 29114038).

Cell types & compartments. CL terms: neuron (CL:0000540), glutamatergic/pyramidal neuron, GABAergic interneuron. Subcellular GO-CC: cytoskeleton/actin cytoskeleton (GO:0015629), dendritic spine (GO:0043197), cytoplasm, membrane (at GTPase docking).

Immune/metabolic involvement. Not a primary feature of IDDMSSD. (Note: much of the broader PAK1 literature concerns oncology/tumor immunity; those roles are not the mechanism of this neurodevelopmental disorder.)

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7. Anatomical Structures Affected

- **Primary organ / system:** the **brain / central nervous system** (UBERON:0000955 brain; UBERON:0001017 CNS). Body system: nervous system.
- **Secondary involvement:** ventricular system (ventriculomegaly, hydrocephalus; UBERON:0002285 brain ventricle), cerebral white matter (UBERON:0002316; white matter

hyperintensities), corticospinal tracts (spastic quadriplegia). Rare non-neural: kidney (horseshoe kidney, UBERON:0002113) in a kinase-domain case.

- **Tissue/cell level:** nervous tissue; neurons (CL:0000540), affecting neurite outgrowth, dendritic arbors, and dendritic spines; synapses.
- **Subcellular:** actin cytoskeleton (GO:0015629), dendritic spine (GO:0043197), cytoplasm.
- **Localization / lateralization:** generally **bilateral** (megalencephaly, ventriculomegaly, diffuse developmental impairment); some structural anomalies may be asymmetric.

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8. Temporal Development

- **Onset:** congenital/early-infancy developmental features; **macrocephaly often postnatal** and progressive during infancy; seizures typically in infancy/early childhood (epileptic spasms in the first year in severe cases). Pattern: insidious/chronic developmental.
- **Progression:** the underlying encephalopathy is **static (non-progressive)**. Epilepsy severity varies and can be medically refractory. No neurodegeneration described.
- **Course / duration:** chronic, lifelong.
- **Remission:** none spontaneous for the neurodevelopmental disability; seizures may be partially controlled with treatment.
- **Critical period:** the postnatal brain-growth window (when PAK1/3-dependent neuronal complexity sets brain size) is the key developmental vulnerability/intervention window (inferred from mouse data, PMID: 21115725).

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9. Inheritance and Population

- **Epidemiology:** ultra-rare; only ~**16–20 patients reported worldwide**. No reliable prevalence/incidence estimates exist.
- **Inheritance pattern: autosomal dominant**, essentially always **de novo** (unaffected parents).
- **Penetrance:** effectively complete in carriers of pathogenic variants (all reported de novo carriers are affected).

- **Expressivity: variable** — from mild DD with normal head circumference to severe ID with refractory epilepsy, spastic quadriplegia, and hydrocephalus (PMID: 36905087; PMID: 41124873).
- **Genetic anticipation:** not applicable (not a repeat-expansion disorder).
- **Mosaicism: post-zygotic mosaicism documented** (p.Leu470Pro, VAF 13.4%); parental germline mosaicism is theoretically possible and underlies a low but non-zero recurrence risk (PMID: 41124873).
- **Founder effects / consanguinity / carrier frequency:** not applicable (de novo dominant; no carrier state).
- **Population demographics:** cases reported across diverse populations (e.g., European, Chinese). No ethnic predilection, geographic clustering, or established sex bias given the small numbers.

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10. Diagnostics

Genetic testing is the diagnostic cornerstone.

- **Recommended approach: trio whole-exome sequencing (WES)** or **whole-genome sequencing (WGS)** to detect the de novo heterozygous missense PAK1 variant and confirm de novo status. This is how all reported cases were identified (PMID: 31504246; PMID: 41124873).
- **Gene panels:** intellectual disability / epilepsy / macrocephaly NGS panels that include PAK1.
- **Single-gene testing:** targeted PAK1 sequencing if the phenotype is highly suggestive.
- **Variant interpretation:** ACMG/AMP criteria; de novo occurrence, absence from gnomAD, domain clustering, and functional gain-of-function evidence support pathogenicity. Mosaic variants require sensitive deep sequencing (low VAF).
- **Chromosomal microarray (CMA) / karyotype:** useful to exclude copy-number contributors (one case had a concurrent 1q44 microdeletion detected by CMA; PMID: 36624491).

Clinical / imaging workup (supportive, non-specific): - **Brain MRI:** megalencephaly, ventriculomegaly, structural anomalies, white matter hyperintensities, hydrocephalus. - **EEG:** for seizure/epilepsy characterization (focal, GTC, hypsarrhythmia in spasms). - **Head circumference monitoring:** documents postnatal macrocephaly. - **Developmental/cognitive assessment:** documents ID and speech delay.

No specific biochemical biomarker, enzyme assay, or omics-based diagnostic test exists for this disorder.

Differential diagnosis: other megalencephaly/macrocephaly–intellectual-disability syndromes, notably **PI3K-AKT-mTOR pathway megalencephaly syndromes** (e.g., PTEN, PIK3CA, AKT3, MTOR), and other Rho-GTPase/actin-regulator neurodevelopmental disorders. Molecular testing distinguishes them.

Screening. No newborn or population carrier screening (de novo dominant). Cascade testing of parents is for recurrence-risk assessment (to detect parental mosaicism), not for at-risk sib detection in the usual sense.

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11. Outcome / Prognosis

- **Survival / mortality:** no disease-specific mortality statistics; the disorder is not characteristically lethal in early life, though severe cases with refractory epilepsy and major structural brain anomalies carry higher morbidity and potential mortality. Life expectancy data are unavailable given rarity.
- **Morbidity / disability:** significant lifelong disability — intellectual disability, speech/communication impairment, epilepsy, and in severe cases motor disability (spastic quadriplegia) requiring full-time care.
- **Disease course: static (non-progressive) encephalopathy**; outcome is largely determined by baseline severity and epilepsy control.
- **Complications:** refractory epilepsy, hydrocephalus requiring shunting, feeding/motor complications in severe cases.
- **Recovery potential:** developmental deficits are permanent; developmental therapies can improve function but not cure.
- **Prognostic factors:** variant domain and severity appear to correlate with outcome — autoregulatory-domain variants associate with more neuroanatomical alterations; kinase-

domain variants with more non-neurological comorbidities (PMID: 36905087). Presence of IESS/refractory epilepsy and major structural anomalies predicts worse outcome.

12. Treatment

There is no approved disease-modifying or targeted therapy. Management is **entirely symptomatic and supportive.**

Modality	Intervention	NCIT (suggested)
Anti-seizure medication	Standard ASMs; vigabatrin / ACTH for epileptic spasms (IESS)	NCIT:C264 (Anticonvulsant)
Neurosurgical	Ventriculoperitoneal shunting for obstructive hydrocephalus	NCIT:C157866 (Shunt placement)
Rehabilitation	Speech therapy, physical therapy, occupational therapy	NCIT:C15312 (Rehabilitation therapy)
Developmental/ educational	Early intervention, special education	—
Supportive	Symptom management, nutrition, orthopedic care	NCIT:C133397 (Supportive care)

Pharmacogenomics: none specific. **Gene/cell/RNA therapy:** none; would face the challenge that the disorder is **gain-of-function**, so an allele-selective silencing (ASO/siRNA) or kinase-inhibition strategy would be mechanistically preferred over gene replacement.

Experimental / rational precision prospect — PAK1 inhibition. Because the pathogenic mechanism is **constitutive PAK1 hyperactivation**, small-molecule **PAK1 inhibitors** (ATP-competitive, allosteric, or degraders) are a mechanistically rational strategy. In a related NDD (fragile X), PAK inhibition (e.g., FRAX486) rescues dendritic-spine/synaptic defects (PMID: 29114038). However, **no PAK1-targeted therapy has been tested in IDDMSSD patients**, and achieving group-I PAK (PAK1 vs PAK2/3) isoform selectivity remains a pharmacological challenge (PMID: 42943325). This remains **preclinical/hypothetical** for this indication.

13. Prevention

- **Primary prevention:** not possible — variants arise de novo; no modifiable risk factors.
- **Secondary prevention:** early molecular diagnosis (trio sequencing) enables timely seizure management and early developmental intervention.
- **Tertiary prevention:** aggressive epilepsy control, hydrocephalus management, and rehabilitation to limit complications and maximize function.
- **Genetic counseling:** recurrence risk for parents of an affected child is **low but non-zero** (parental germline mosaicism possible); an affected individual, if reproductive, would have 50% transmission risk (autosomal dominant). Prenatal/preimplantation testing is feasible for a known familial variant.
- **Immunization / public health / environmental interventions:** not applicable.

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14. Other Species / Natural Disease

- **Orthologous genes:** mouse *Pak1* (NCBI Gene 18479; MGI), rat *Pak1* (RGD), zebrafish *pak1* (ZFIN), *Drosophila Pak/dPak* (FlyBase), *C. elegans pak-1* (WormBase). PAK1 is **highly evolutionarily conserved**.
- **Natural disease in other species:** no naturally occurring ortholog of this specific human disorder is reported in companion animals or wildlife (OMIA); not a veterinary disease of note.
- **Comparative biology:** the Rac/Cdc42 → PAK → cofilin actin-regulatory mechanism is deeply conserved across metazoans, making invertebrate and vertebrate models informative for mechanism.
- **Transmission / zoonosis:** not applicable (genetic disorder).

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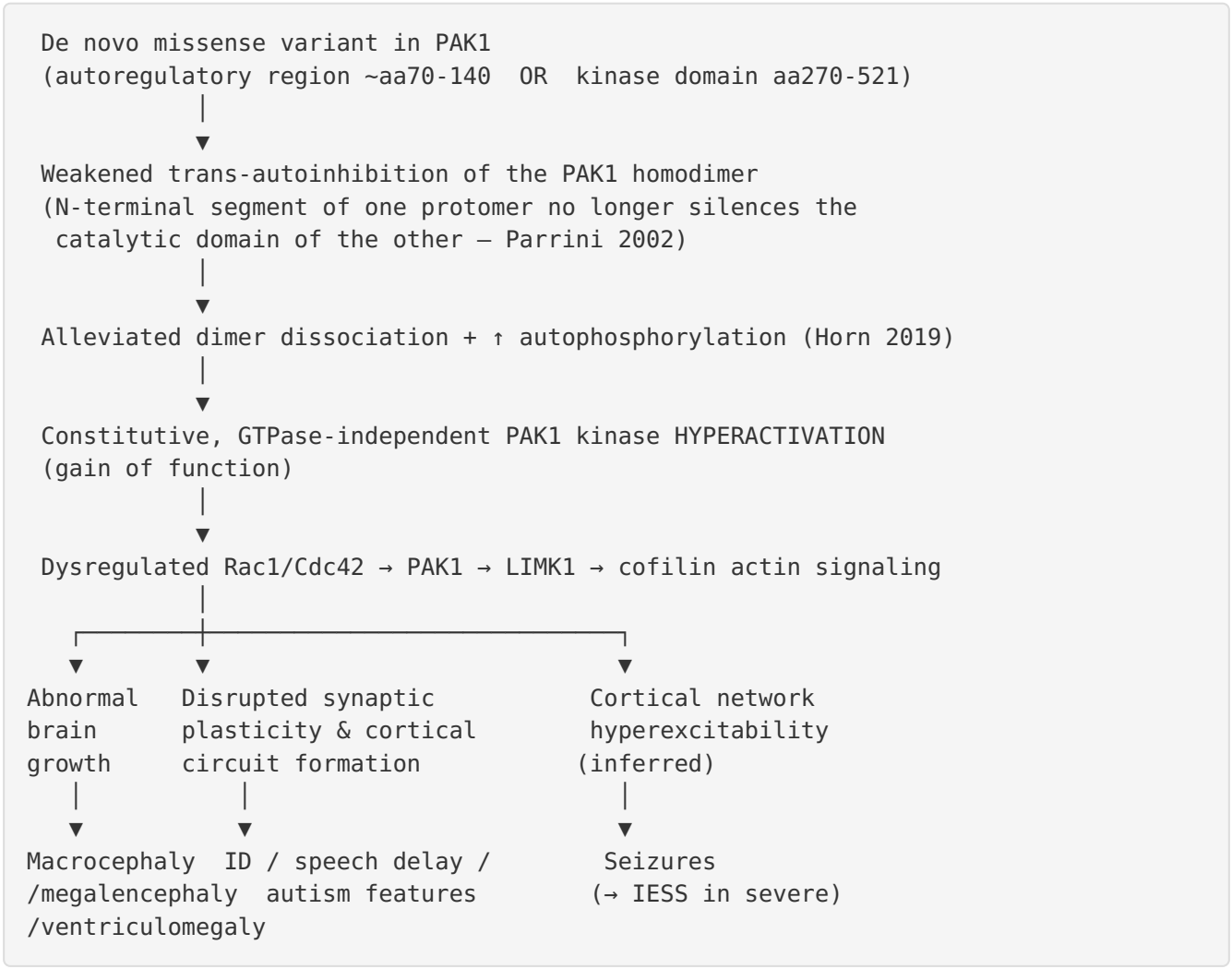
15. Model Organisms

- **Mouse (primary CNS model):** *Pak1*-null mice are **viable and fertile** (with immune/vascular phenotypes). The key neurodevelopmental insight comes from ***Pak1/Pak3* double-knockout mice**, which show severely impaired postnatal brain growth, reduced brain

volume, increased cell density, simplified dendrites/axons, reduced synapse density, and impaired bidirectional synaptic plasticity, with altered cofilin activity (PMID: 21115725).

- **Other models:** zebrafish, *Drosophila*, and *C. elegans* orthologs exist for mechanistic studies of Rho-GTPase/PAK/actin signaling.
- **Genetic model types available:** knockout (Pak1, Pak1/Pak3 DKO). **A patient-matched gain-of-function knock-in rodent** that recapitulates IDDMSSD macrocephaly/seizures is **not yet reported** in the retrieved literature.
- **Phenotype recapitulation:** existing KO models demonstrate PAK1's requirement for brain growth and synaptic function but model **loss of function** — the *opposite direction* of the disease's gain-of-function mechanism. This is the principal model limitation: a constitutively-active knock-in is needed to faithfully model IDDMSSD.
- **In vitro:** cell-based functional assays measuring dimer dissociation, autophosphorylation, kinase activity, and neurite actin dynamics were used to demonstrate the gain-of-function mechanism of patient variants (PMID: 31504246).

Mechanistic Model / Interpretation



Domain-dependent genotype–phenotype axis:

Variant location	Predominant phenotype tendency
Autoregulatory domain (~aa 70–140)	More **neuroanatomical** alterations
Protein kinase domain (aa 270–521)	More **non-neurological** comorbidities

(Per Corriveau et al. 2023: *"neuroanatomical alterations were found more often in individuals with PAK1 variants in the autoregulatory domain. In contrast, non-neurological comorbidities were found more often in individuals with PAK1 variants in the protein kinase domain"*, PMID: 36905087.)

The unifying theme is that **both variant classes converge on the same molecular outcome — relief of autoinhibition and constitutive kinase activity** — but their differing structural contexts tune the clinical presentation. The all-missense spectrum and the gnomAD constraint profile (mis_Z 5.50, pLI 0.07) are mutually reinforcing lines of evidence for gain of function: the gene cannot tolerate missense change (because such changes activate it pathogenically) yet tolerates loss-of-function alleles (because simple haploinsufficiency does not cause this disease).

Evidence Base

PMID	Title (abbrev.)	Contribution
31504246	*De novo variants in PAK1 lead to ID with macrocephaly and seizures*	**Founding gene–disease paper**; 4 de novo missense; functional GoF (dimer dissociation, autophosphorylation, actin dynamics)
41124873	*Novel PAK1 variants: variable spectrum mild DD → IESS*	5 new variants + literature review; **all 16 variants missense**; epilepsy/macrocephaly frequencies; documented mosaicism
36905087	*PAK1 p.Leu470Gln: epilepsy, macrocephaly, spastic quadriplegia, hydrocephalus*	Common features list; **domain-dependent genotype–phenotype**; first recurrent kinase-domain residue (Leu470)
36624491	*De novo PAK1 (p.T84R) + 1q microdeletion*	Autoregulatory/CRIB variant; phenotype attributed to PAK1; CMA value
11804587	*Pak1 homodimers autoinhibited in trans; dissociated by Cdc42/Rac1*	**Structural basis** of autoinhibition disrupted by disease variants
21115725	*PAK1 & PAK3 control brain size*	**Mouse genetics**: PAK1/3 required for postnatal brain growth; cofilin-actin link
29114038	*Aberrant Rac1-cofilin signaling in fragile X*	Validates Rac-PAK-cofilin axis as pathogenic in NDD; **rationale for PAK inhibition**
42943325	*Targeting PAK1 within group I PAKs*	PAK1-inhibitor pharmacology; isoform-selectivity challenges
19890013	*IP3K-A scaffolds synaptic Rac signaling*	Context on Rac→PAK1 synaptic actin regulation

Evidence source types: **human clinical/genetic** (31504246, 41124873, 36905087, 36624491); **in vitro functional** (31504246, 11804587); **model organism** (21115725, 29114038).

Limitations and Knowledge Gaps

1. **Ultra-small sample size (~16–20 patients).** Frequencies, genotype–phenotype correlations, and prognosis are based on case reports/series and are provisional.
2. **No patient-matched gain-of-function animal model.** Available mouse models are loss-of-function (opposite mechanistic direction); the disease's GoF biology has not been modeled in vivo.
3. **Epileptogenic mechanism is inferred,** not directly demonstrated for patient variants — the link from PAK1 hyperactivation to cortical hyperexcitability remains to be shown.
4. **No validated biomarker** or omics signature for diagnosis, severity, or treatment response.
5. **No therapeutic data in patients** — PAK1 inhibition is rational but untested in this indication, with unresolved isoform-selectivity and CNS-delivery challenges.
6. **Incomplete ontology coverage** (Orphanet/ICD/MeSH lack a dedicated specific term), complicating epidemiological capture.
7. **Natural history is sparsely documented** — no longitudinal cohorts.

Proposed Follow-up Experiments / Actions

1. **Generate a conditional, patient-variant knock-in mouse** (e.g., Pak1 p.Leu470 or an autoregulatory-domain variant) to test whether constitutive activation recapitulates megalencephaly, seizures, and behavioral deficits — the single most important missing model.
2. **iPSC-derived neurons / cerebral organoids** from patients (or isogenic CRISPR knock-in lines) to quantify neurite/spine phenotypes, network excitability (MEA), and brain-organoid overgrowth.
3. **Preclinical PAK1-inhibitor rescue studies** in the above models (patient neurons/organoids and knock-in mice), benchmarking efficacy and isoform selectivity, to establish proof-of-concept for a precision therapy.
4. **International patient registry / natural-history study** to refine phenotype frequencies, domain-specific genotype–phenotype correlations, seizure outcomes, and prognosis.
5. **Structural / biochemical characterization** of each recurrent variant (dimer stability, activation kinetics) to build a variant-level functional map and improve ACMG classification of future VUS.

6. **Deep-sequencing mosaicism screening** in parents of probands to quantify germline/somatic mosaicism and refine recurrence-risk counseling.
7. **Allele-selective silencing (ASO/siRNA) feasibility** targeting the mutant PAK1 allele as an alternative GoF-directed therapeutic strategy.

Report compiled from a 5-iteration autonomous investigation; 9 confirmed findings; 14 papers reviewed. All mechanistic and clinical claims are cited to primary literature (PMID). Where evidence is absent or inferential, this is stated explicitly.
